SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Problems

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Problems
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:		**Date:**	30/10/2026

Code of Conduct

I am Renuga Devi B(192311111) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renuga Devi B

Instructions

- This test contains 20 questions.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log 10$ of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given:

- Total spectrum = 45 MHz = 45,000 kHz
- Bandwidth per simplex channel = 25 kHz

A full-duplex channel requires two simplex channels.

Formula:

Full-duplex channels = Total bandwidth ÷ (2 × Simplex channel bandwidth)

Calculation:

$$N = \frac{45{,}000}{2\times25} = \frac{45{,}000}{50} = 900$$

Answer: 900 full-duplex voice channels.

2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given:

- Total channels = 560
- Cluster size = 7 cells

Formula:

$$Channelspercell = \frac{Totalchannels}{Clustersize}$$

Calculation:

$$N = \frac{560}{7} = 80$$

Answer: 80 channels per cell.

3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given:

- Number of carriers = 100
- Time slots per carrier = 8

Formula:

Total voice channels = Carriers × Time Slots

Calculation:

$$N = 100 \times 8 = 800$$

Answer: 800 simultaneous voice channels.

4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Given:

- Base station power = 80 W
- Handset power = 4 W

Formula:

$$P_{dB} = 10\log_{10}\left(\frac{P_1}{P_2}\right)$$

Calculation:

$$P_{dB} = 10\log_{10}\left(\frac{80}{4}\right)$$

$$= 10\log_{10}(20)$$

$$= 10(1.3010) = 13.01dB$$

Answer: 13.01 dB.

5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given:

- Carrier power = −60 dBm
- Interference power = −92 dBm

Formula:

$$C/I = P_C - P_I$$

Calculation:

$$C/I = (-60) - (-92)$$

$$= -60 + 92 = 32dB$$

Answer: 32 dB.

6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Given:

- Total spectrum = 42 MHz
- Bandwidth per channel = 300 kHz
- Cluster size = 7

Step 1: Calculate total channels.

$$N = \frac{42{,}000}{300} = 140$$

Step 2: Calculate channels per cell.

$$N_{cell} = \frac{140}{7} = 20$$

Answer: 20 channels per cell.

7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given:

- Initial received power = 64 W
- Distance increases by a factor of 2
- Path-loss exponent $n = 4$

Formula:

$$P_2 = P_1\left(\frac{d_1}{d_2}\right)^n$$

Since $d_2 = 2d_1$,

$$P_2 = 64\left(\frac{1}{2}\right)^4$$

$$P_2 = \frac{64}{16} = 4W$$

Answer: 4 W.

8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Given:

- New radius = 2 × Original radius
- Path-loss exponent $n = 2$
- SNR must remain constant

Formula:

$$\frac{P_2}{P_1} = \left(\frac{d_2}{d_1}\right)^n$$

Calculation:

$$\frac{P_2}{P_1} = 2^2 = 4$$

Answer: Transmit power must increase by a factor of 4 (four times the original power).

9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given:

- Cluster size = 4 cells
- Channels per cell = 15

Formula:

Total Clusters = Cells × Channels per cell

Calculation:

$$N = 4 \times 15 = 60$$

Answer: 60 channels per cluster.

10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given:

- Vehicle speed = 100 km/h
- Carrier frequency = 900 MHz
- Speed of light $c = 3 \times 10^8 m/s$

Step 1: Convert speed to m/s.

$$v = \frac{100 \times 1000}{3600} = 27.78 m/s$$

Formula:

$$f_D = \frac{v f_c}{c}$$

Assuming the vehicle moves directly toward or away from the transmitter

Calculation:

$$f_D = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8}$$

$$f_D = 83.34Hz$$

Answer: Approximately 83.3 Hz.

11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given:

- Frame duration = 5 ms
- Number of slots = 8

Formula:

$$Slotduration = \frac{Frameduration}{Numberofslots}$$

Calculation:

$$T = \frac{5}{8} = 0.625ms$$

Answer: 0.625 ms per slot.

12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given:

- Total bandwidth = 24 MHz
- Carrier spacing = 200 kHz

Formula:

$$N = \frac{Totalbandwidth}{Carrierspacing}$$

Calculation:

$$N = \frac{24{,}000}{200} = 120$$

Answer: 120 carrier frequencies.

13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given:

- Transmitted power = 8 W
- Path loss = 33 dB

Step 1: Convert transmitted power into dBm.

$$P_{dBm} = 10\log_{10}(P_{mW})$$

$$8W = 8000mW$$

$$P_{dBm} = 10\log_{10}(8000)$$

$$P_{dBm} = 39.03dBm$$

Step 2: Calculate received power.

$$P_R = P_T - Pathloss$$

$$P_R = 39.03 - 33$$

$$P_R = 6.03dBm$$

Answer: 6.03 dBm.

14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Given:

- Frequency = 1900 MHz
- Distance = 2.5 km

Formula:

For frequency in MHz and distance in km,

$$FSPL = 32.44 + 20\log_{10}(f) + 20\log_{10}(d)$$

Calculation:

$$FSPL = 32.44 + 20\log_{10}(1900) + 20\log_{10}(2.5)$$

$$= 32.44 + 65.58 + 7.96$$

$$FSPL = 105.98dB$$

Answer: Approximately 105.98 dB.

15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given:

- Number of cells = 6
- Capacity per cell = 250 calls

Formula:

Total capacity =Number of cells × Capacity per cell

Calculation:

$$N = 6 \times 250 = 1500$$

Answer: 1,500 simultaneous calls.

16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Given:

- Call arrival rate = 4000 calls/second
- Average call duration = 3 minutes = 180 seconds

Formula (Little's Law):

$$N = \lambda T$$

Where:

- λ = arrival rate
- T = average time spent in the system

Calculation:

$$N = 4000 \times 180$$

$$N = 720,000$$

Answer: 720,000 active calls

17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given:

- Number of channels = 150
- Bandwidth per channel = 25 kHz

Formula:

$$B = N \times B_{channel}$$

Calculation:

$$B = 150 \times 25 = 3750kHz$$

$$B = 3.75MHz$$

Answer: 3.75 MHz.

18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given:

- Number of adjacent frequency bands = 30
- Guard band between each adjacent pair = 40 kHz

Step 1: Calculate the number of gaps.

For 30 bands arranged in a row:

$$Numberofgaps = 30 - 1 = 29$$

Step 2: Calculate total guard-band bandwidth.

$$B_G = 29 \times 40$$

$$B_G = 1160kHz$$

$$B_G = 1.16MHz$$

Answer: 1.16 MHz.

19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Given:

- Vehicle speed = 90 km/h
- Cell radius = 1.5 km
- Cell diameter = 3 km

Step 1: Calculate the distance travelled in one hour.

$$D = 90km$$

Step 2: Calculate the approximate number of cell crossings.

$$N = \frac{Distancetravelled}{Celldiameter}$$

$$N = \frac{90}{3} = 30$$

Answer: Approximately 30 handoffs per hour.

20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given:

- Regional centres = 8
- Base stations per regional centre = 40
- Users per base station = 250

Formula:

$$N = 8 \times 40 \times 250$$

Calculation:

$$8 \times 40 = 320$$

$$320 \times 250 = 80,000$$

Answer: 80,000 users.

21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Given:

- Total area = 400 km²
- Cell radius = 400 m = 0.4 km
- Approximate cell area = R^2

Step 1: Calculate the area of one cell.

$$A = R^2$$

$$A = (0.4)^2 = 0.16km^2$$

Step 2: Calculate the number of cells.

$$N = \frac{Totalarea}{Areapercell}$$

$$N = \frac{400}{0.16} = 2500$$

Answer: Approximately 2,500 cells.

22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given:

- Total spectrum = 6 MHz
- Bandwidth per channel = 30 kHz

Formula:

$$N = \frac{Totalbandwidth}{Bandwidthperchannel}$$

Calculation:

$$N = \frac{6000}{30} = 200$$

Answer: 200 channels.

23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given:

- Number of subscribers = 1200
- Calls per subscriber per hour = 4
- Average call duration = 2.5 minutes

Formula:

$$A = \frac{Totalcalldurationperhour}{3600}$$

First, calculate the total calls per hour:

$$1200 \times 4 = 4800 calls/hour$$

Convert the average call duration to seconds:

$$2.5 \times 60 = 150 seconds$$

Total call duration per hour:

$$4800 \times 150 = 720,000 seconds$$

Calculate offered traffic:

$$A = \frac{720,000}{3600} = 200$$

Answer: 200 Erlangs.

24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Given:

- Total bandwidth = 40 MHz
- Bandwidth per user = 400 kHz

Formula:

$$N = \frac{Totalbandwidth}{Bandwidthperuser}$$

Calculation:

$$N = \frac{40,000}{400} = 100$$

Answer: 100 simultaneous users.

25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given:

- Number of base stations = 1200
- Capacity per base station = 600 users

Formula:

Total capacity = Number of base stations × Users per base station

Calculation:

$$N = 1200 \times 600$$

$$N = 720,000$$

Answer: 720,000 simultaneous users.